

DDR1 vs DDR2 SDRAM

Memory Architecture Evolution

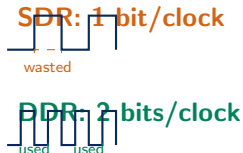


JESD79 & JESD79-2 Standards

Motivation: Why DDR Memory?

SDR SDRAM Limitation:

- Data valid on **rising edge only**
- 50% bus utilization (falling edge wasted)
- Frequency scaling hit thermal/power limits
- Needed way to increase bandwidth without more MHz



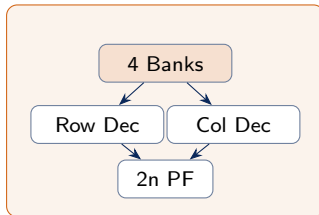
Solution: Capture on Both Clock Edges

Transfer data on rising **AND falling edges** of the same clock → doubles bandwidth without doubling frequency.

DDR1 (JESD79): Architecture Overview

Key Features:

- **2:1 Prefetch** → 2 bits per cell per access
- **4 Banks** with independent row buffers
- **VDD = 2.5 V**, SSTL_2 I/O standard
- **DQS edge-aligned** to data (requires center logic)
- **No ODT** (external termination only)



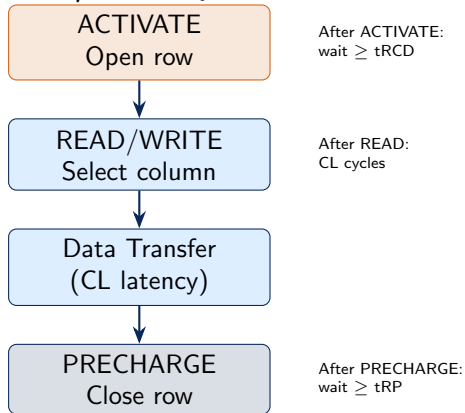
Differential Clock (CK / CK#)

Speed Grades

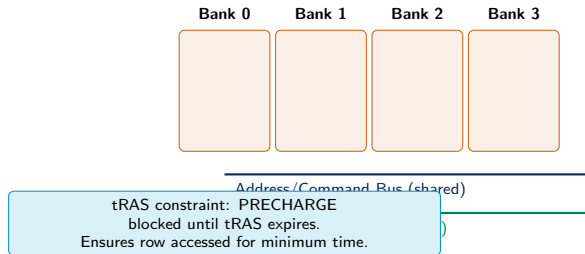
DDR-200 (10 ns), DDR-266 (7.5 ns),
DDR-333 (6 ns), DDR-400 (5 ns)

DDR1: Command Sequence (ACTIVATE → READ/WRITE → PRECHARGE)

Read/Write Operation:



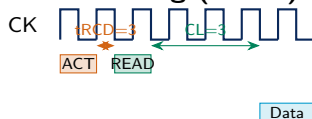
4-Bank Architecture:



DDR1: Key Timing Parameters

Parameter	Typical (DDR-400)
CL	3 cycles (15 ns)
tRCD	3 cycles (15 ns)
tRP	3 cycles (15 ns)
tRAS	8 cycles (40 ns)
tWR	2 cycles
tRFC	12 cycles (72 ns)

READ Timing (CL=3):



CL Cycles from READ command to data valid

tRCD Cycles from ACTIVATE to READ/WRITE

tRP Cycles to fully close a row

Technical Challenges:

- **DQS Edge-Aligned:** Controller must perform internal centering → complex logic
- **Only 4 Banks:** Limited interleaving → poor parallelism
- **2.5 V Supply:** High power consumption; thermal constraints
- **No ODT:** External PCB termination required; SI issues at higher speeds
- **Frequency Hard Ceiling:** Capped at 400 MT/s

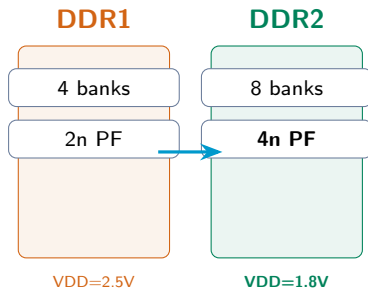
Design Trade-offs:

- Simple command flow
- Low latency (CL=2-3)
 - × Difficult DQ centering
 - × Power per bit increasing
 - × Speed scaling limited

DDR2 (JESD79-2): Architectural Improvements

Major Enhancements:

- **4:1 Prefetch** → 4 bits per cell per access
- **8 Banks** → double parallelism vs DDR1
- **VDD = 1.8 V** → 28% lower power
- **ODT On-Die** → controller-driven, no PCB resistors
- **DQS Center-Aligned** → easier capture, no internal centering
- Data rates: 400–1066 MT/s



On-Die Termination (ODT):

- Resistors built **inside the DRAM**
- Controller enables via **ODT signal**
- Values: $50\ \Omega$, $75\ \Omega$, $150\ \Omega$ (set in EMR)
- Reduces reflections → better SI at speed

8 Banks:

- Opens more rows in parallel
- Better command interleaving
- Requires new constraint: **tFAW** (Four Activate Window)

Additive Latency (AL):

READ command can be issued early (at ACTIVATE time) → DRAM queues internally.

No AL	With AL=1
ACT → wait tRCD	ACT (at cycle 0)
READ (at +tRCD)	READ (at cycle 1)
Data at tRCD+CL	Data at AL+CL (same absolute time)

→ Better bus utilization, pipelined commands

DDR2: Timing Concepts – RL, WL, CWL

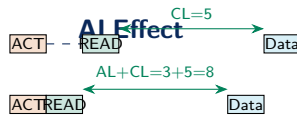
Read/Write Latencies:

Term	Definition
RL (Read Latency)	$AL + CL$
WL (Write Latency)	$AL + CWL$
CWL (CAS Write Latency)	Separate from CL
tRCD	RAS-to-CAS delay

Example (DDR2-800):

- $CL = 5$ cycles, $AL = 0$ or 3
- $CWL = 5$ cycles
- $RL = 5$ ($AL=0$) or 8 ($AL=3$)

AI Effect on Timing:



Both reach data at same time,
but commands posted earlier with AL.

DDR1 vs DDR2: Comprehensive Comparison

Feature	DDR1	DDR2
Prefetch	2n	4n
Banks	4	8
VDD	2.5 V	1.8 V
DQS Alignment	Edge	Center
ODT	External	On-die
Additive Latency	No	Yes (AL)
CWL (Write)	CL-1	Separate

Key Takeaways

DDR1 Strengths

- ✓ Simpler architecture
- ✓ Lower latency (CL 2–3)
- ✓ Easier to design/test
- ✓ Proven in high-volume

DDR1 Limits

- × Only 4 banks
- × 2.5 V power loss
- × DQS centering complexity
- × Speed bottleneck

DDR2 Advances

- ✓ 8 banks for parallelism
- ✓ ODT simplifies SI design
- ✓ 1.8 V cuts power
- ✓ Higher bandwidth

DDR2 Trade-offs

- × Higher CL (clock cycles)
- × AL adds complexity
- × More signals/timing rules
- × Tighter SI requirements

Memory Evolution

DDR2 trades latency for bandwidth & efficiency.

Summary

DDR1 (2005)

- 2:1 Prefetch, 4 banks
- 2.5 V SSTL_2 I/O
- DQS edge-aligned
- CL: 2–3 cycles
- No ODT/AL/tFAW
- Up to 400 MT/s
- DFI 1.0

DDR2 (2009)

- 4:1 Prefetch, 8 banks
- 1.8 V SSTL_18 I/O
- DQS center-aligned
- CL: 3–7 cycles
- ODT, AL, tFAW added
- Up to 1066 MT/s
- DFI 2.1

DDR2: $2.7\times$ bandwidth, 28% lower power than DDR1

Key: more banks, on-die termination, posted commands